**Name:** **Roll Number:**

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## 1. Heap Implementations – 30 Marks

### A. Binary Heap Implementation (10 marks)

| Criteria | Marks |
|---|---|
| Correct implementation of insert, extract min, find min, decrease key operations | 9 |
| Code clarity and modularity | 1 |

### B. Fibonacci Heap Implementation (10 marks)

| Criteria | Marks |
|---|---|
| Correct implementation of insert, extract min, find min, decrease key operations | 6 |
| Cascading cut implementation | 3 |
| Code clarity & correct node structure management | 1 |

### C. Hollow Heap Implementation (10 marks)

| Criteria | Marks |
|---|---|
| Correct implementation of insert, decrease-key using hollow nodes, extract-min | 9 |

| | |
|---|---|
| Code clarity & correct node structure management | 1 |

**2. Dijkstra Integration – 15 Marks**

**A. Priority Queue Interface Integration (5 marks)**

| **Criteria** | **Marks** |
|---|---|
| Clean abstraction allowing plug-and-play heap usage | 10 |

**B. Dijkstra Implementation (10 marks)**

| **Criteria** | **Marks** |
|---|---|
| Correctness of shortest path computation | 5 |

**3. Performance Measurement & Data Recording – 20 Marks**

**A. Operation Time Measurement (8 marks)**

| **Criteria** | **Marks** |
|---|---|
| Accurate timing mechanism & multiple-run averaging | 3 |
| Comparison across all three heaps | 5 |

**B. Heap Structural Statistics (6 marks)**

| **Criteria** | **Marks** |
|---|---|
| Recording height, number of trees, cascading cuts (Fibonacci) | 6 |

**C. Total Runtime Measurement for Dijkstra (4 marks)**

| Criteria | Marks |
|---|---|
| Per-heap total execution time | 4 |

**D. Memory Usage Measurement (2 marks)**

| Criteria | Marks |
|---|---|
| Recording and comparing memory footprint of all heaps | 2 |

**4. Experimental Setup & Execution – 15 Marks**

**A. Dataset Loading & User File Selection System (5 marks)**

| **Criteria** | **Marks** |
|---|---|
| Prompts user to select dataset & correct graph parsing | 5 |

**B. Experiment A: Static Routing (5 marks)**

| **Criteria** | **Marks** |
|---|---|
| Correct execution on all datasets | 5 |

**C. Experiment B: Random Operation Profiling (5 marks)**

| **Criteria** | **Marks** |
|---|---|
| Correct generation of 100,000 operations | 3 |
| Valid performance results | 2 |

**5. Report Quality & Analysis – 20 Marks**

**A. Theoretical Overview (5 marks)**

| **Criteria** | **Marks** |
|---|---|
| Correct explanation of heap concepts, complexities, behavior | 10 |

**B. Results Presentation – Tables & Plots (5 marks)**

| Criteria | Marks |
|---|---|
| Clear tables & readable graphs with correct units | 3 |
| Side-by-side comparisons | 2 |

**C. Analytical Insights & Discussion (5 marks)**

| Criteria | Marks |
|---|---|
| Explanation of why certain heaps performed better | 3 |
| Discussion of trade-offs (speed, memory, complexity) | 2 |

**Bonus – Up to 10 Marks (Optional)**

**A. Parallelized Dijkstra implementation**

| Criteria | Marks |
|---|---|
| Bonus marks | 5 |

**B. Heap evolution visualization**

| Criteria | Marks |
|---|---|
| Bonus marks | 5 |